

PUSALA PREETHAM

Software Developer | B.Tech Computer Science and Engineering (Machine Learning & Full-Stack)

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SUMMARY

Computer Science undergraduate (GPA 9.33/10) with hands-on experience in full-stack development and applied machine learning. Built a multi-horizon electricity demand forecasting pipeline (R^2 up to 0.93) during a research internship, and shipped full-stack apps using React, Node.js, and Flask. Solved 550+ DSA problems on LeetCode and CodeChef. Seeking a Software Engineering internship to apply strong CS fundamentals and research-driven problem-solving.

EDUCATION

SRM University, Amaravati

2024 – 2028

B.Tech in Computer Science and Engineering

Andhra Pradesh, India

- Cumulative GPA: 9.33 / 10
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management, Computer Networks, Operating Systems

EXPERIENCE

Research Intern — Electricity Consumption Forecasting using ML & DL

Summer 2026

Applied machine learning research internship

- Built a multi-horizon (1-hour, 1-day, 1-month) electricity consumption forecasting pipeline using XGBoost, LightGBM, and CatBoost on 3-minute-interval smart meter data from 38 households (2019–2021).
- Achieved R^2 scores of 0.93 (1-day) and 0.92 (1-month) with CatBoost, and 0.80 (1-hour) with LightGBM, benchmarked using MAE, RMSE, and SMAPE across all three models.
- Engineered rolling-statistics, EMA, trend, and time-based features, and integrated live weather data (temperature, humidity, precipitation, wind speed, surface pressure) via the Open-Meteo API to improve model accuracy.
- Ran feature importance analysis to interpret model behavior and document findings using Python, Pandas, and Matplotlib.

PROJECTS

Electricity Consumption Forecasting Pipeline

Technologies: Python, XGBoost, LightGBM, CatBoost, Pandas, Scikit-learn, PyArrow, Open-Meteo API

- Designed and implemented an end-to-end forecasting pipeline predicting household electricity consumption across three horizons (hourly, daily, monthly).
- Transformed raw 3-minute-interval smart meter readings into model-ready datasets, engineering 20+ features including rolling statistics, EMA, and trend indicators.
- Trained and benchmarked three gradient boosting models, with CatBoost achieving the best long-horizon performance ($R^2 = 0.93$ daily, 0.92 monthly).

AI-Based Student Dropout Prediction System

[GitHub](#)

Technologies: Node.js, HTML, CSS, JavaScript, Python, Flask, Python ML libraries

- Developed a machine learning model to predict student dropout risk from academic and behavioral data.
- Performed data preprocessing and feature selection to improve model accuracy and reduce noise.
- Implemented and compared Logistic Regression, Random Forest, and XGBoost classifiers, selecting the best-performing model by accuracy and precision.

Sports Tournament Management System

[GitHub](#)

Technologies: Node.js, Express.js, MySQL, HTML, CSS, JavaScript, Firebase Authentication

- Built a full-stack web application to manage sports tournaments, including team creation, scheduling, and match management.
- Designed a MySQL relational database schema to store teams, players, matches, and tournament results.
- Developed RESTful APIs using Node.js and Express.js, with Firebase Authentication for secure sign-up, login, and role-based access.

TECHNICAL SKILLS

- Languages: Python, JavaScript, C, SQL
- Web: React, Tailwind CSS, HTML, CSS (Frontend) · Node.js, Express.js
- Databases & Tools: PostgreSQL, MySQL, MongoDB, Git, GitHub

CODING PROFILES

- LeetCode: 367+ problems solved — leetcode.com/u/preetham_pusala
- CodeChef: 194 problems solved — codechef.com/users/preetham1703

ACHIEVEMENTS & ACTIVITIES

- Participated in VibeCraft (HackSRM 7.0), VoyageHack 3.0, and the Sustainable Development Goals Hackathon.